

# BADR ALDEEN AL-KHAZAN

Industrial Engineering, Operations Research & Decision Analytics, Supply Chain & Sustainable Systems

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## PROFESSIONAL SUMMARY

Final-year CAPM-certified Industrial Engineering student (GPA 3.90/4.00) focused on applied Operations Research and Decision Analytics for supply chain, manufacturing, and sustainable systems. Experience spans simulation, optimization, queueing, stochastic modelling, predictive maintenance, and data-driven decision support, with active research and publication experience.

## EDUCATION

**B.Sc. Industrial Engineering (International Program)**, Universitas Islam Indonesia *Expected 2027*

**GPA:** 3.90/4.00. **Relevant coursework:** Operations Research, Production Planning, Simulation & Modelling, Facility Layout, Six Sigma Quality, Manufacturing Processes.

## EXPERIENCE & LEADERSHIP

**Simulation & Modelling Trainee**, Delsim Laboratory, Universitas Islam Indonesia *Jul 2025 – Sep 2025*

- Built and validated discrete-event and system-dynamics models using FlexSim and PowerSim to analyze industrial system behavior.

**Deputy Head, Events Division & Industrial Engineering Ambassador**, UII Global (Volunteer) *Jul 2025 – Dec 2025*

- Led academic and cultural events, logistics and partnerships; served as PIC for international-student activities.

**Deputy Head, Social Responsibility Circle**, Yemeni Students' Union in Indonesia

- Coordinated student-support and community programs and major cultural events.

## RESEARCH & PUBLICATIONS

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| Published    | <b>How Depending on AI Can Impact the Analytical Skills and Critical Thinking of Young People</b> , <i>UII Repository</i> , 2026                                   |
| Published    | <b>Germany's DIGITAL Cooperative Framework: Advancing Sustainable, Inclusive, and Innovative Global Digital Infrastructure</b> , <i>UII Repository</i> , 2025      |
| Accepted     | <b>Simulation-Based Analysis of the Bullwhip Effect in a Multi-Echelon Bottled Water Supply Chain</b> , <i>Accepted for publication; in production</i> , 2026      |
| Conference   | <b>Calibration and Cost-Sensitive Threshold Selection for Imbalanced Predictive Maintenance: an AI4I 2020 Study</b> , <i>Accepted for presentation</i> , TMIC 2026 |
| Under Review | <b>Supply Chain Governance in Sustainability Reporting: a SCOR Orchestrate Analysis of an Indonesian State-Owned Enterprise</b> , 2026                             |
| Under Review | <b>Reducing University Motorcycle Exit Time through Automated Parking Access Control (SIGAP)</b> , 2026                                                            |
| Under Review | <b>Teaching Sustainability with Unsustainable Tools: Toward a Reflexive Green-AI Literacy Framework for Education for Sustainable Development</b> , 2026           |
| In Progress  | <b>Green Vehicle Routing for Electric Truck Distribution with Load-Dependent Energy Consumption</b> , 2026                                                         |

## AWARDS & CERTIFICATIONS

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| <ul style="list-style-type: none"><li><b>1st Place:</b> EXPO PSIT 2026, SIGAP RFID Automated Parking Access System</li><li><b>Best Presentation:</b> EXPO RSKE 2025, Multifunctional Ergonomic Pen</li><li><b>Future Global Leaders Scholarship:</b> Universitas Islam Indonesia (2023–2027)</li><li><b>TOEFL iBT 90/120:</b> ETS (2026–2028)</li></ul> | <ul style="list-style-type: none"><li><b>1st Place:</b> Idea Champion 2.0, AI-PHL Guardian</li><li><b>Outstanding Delegate:</b> IYEC 12, Malaysia &amp; Singapore</li><li><b>CAPM®:</b> Project Management Institute (2026–2029)</li><li><b>Aspire Leaders Program:</b> Aspire Institute (2025)</li></ul> |
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## SKILLS & LANGUAGES

**Technical:** FlexSim, PowerSim, simulation, queueing analysis, stochastic modelling, supply chain analytics, SPC & Six Sigma, production planning, facility layout, work measurement, database design, Advanced Excel, technical writing.

**Professional:** Project management, analytical thinking, team leadership, cross-cultural communication, event management, public speaking.

**Languages:** Arabic (Native), English (B2; TOEFL iBT 90), Indonesian (B2), German (A1).